Ain Shams University – Faculty of Engineering I-CHEP

CSE356: Internet of Things

Project Report

Smart Mosque Management System

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Submitted to	Prof. Islam Tharwat
	Eng. Hamdy Osama

Submitted by	23P0184 – Sara Ahmed
	23P0360 – El-Sharqawy Wael
	23P0386 – Esraa Mustafa
	23P0272 – Nada Medhat

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1. Purpose & Requirements

1.1 Purpose of the IoT System

The purpose of the Smart Mosque Management System is to design an intelligent IoT-based solution that enhances the operational efficiency of mosques by:

- Optimizing energy consumption (lighting and air conditioning)
- Conserving water usage in wudu (ablution) areas
- Monitoring and managing crowd occupancy
- Improving indoor air quality and comfort

The system leverages real-time sensor data, automated control mechanisms, and a centralized monitoring dashboard to ensure efficient resource utilization, especially during high-attendance periods such as Friday prayers and Ramadan.

1.2 Functional Requirements

These define what the system must do.

A. Energy Management

- Automatically turn ON/OFF lights and fans/AC based on:
  - Prayer times (via API)
  - Occupancy detection (motion sensors)
- Adjust energy usage dynamically depending on crowd density.
- Smart window control: automatically opens for natural ventilation when AC is off and humidity $\geq 60\%$; closes when AC activates to retain conditioned air.
- Humidity comfort control: humidifier activates when indoor humidity drops below 30%.

B. Occupancy Detection

- Count number of people entering/exiting using PIR sensors.
- Provide real-time hall capacity status.
- Activate a dedicated siren (is-mosque-full-siren) when all three prayer hall PIR sensors are simultaneously active, indicating maximum occupancy.

C. Water Monitoring (Wudu Area)

- Measure water flow using flow sensors.
- Detect abnormal usage (leaks or taps left open).
- Automatically close water supply using water valves.

D. Air Quality Monitoring

- Measure air quality (e.g., CO_2, humidity, temperature).
- Trigger ventilation (fans) when thresholds are exceeded.

E. Central Dashboard

- Display:
  - Current occupancy
  - Energy consumption
  - Water usage
  - Air quality readings
- Allow manual override control for administrators.

F. Notifications & Alerts

- Send alerts for overcrowding, water leakage, poor air quality, and system faults.

1.3 Non-Functional Requirements

These define how well the system should perform.

A. Performance

- Real-time response (low latency in sensor-actuator loop)
- Efficient handling of multiple sensors simultaneously

B. Scalability

- System should support small to large mosques
- Easy addition of new sensors/devices

C. Reliability

- Continuous operation with minimal downtime
- Fault-tolerant communication between devices

D. Security

- Encrypted communication using WPA2-PSK for wireless sensor-to-gateway connections
- Dashboard access control with username/password authentication
- MAC filtering to allow only authorized IoT nodes

E. Usability

- Simple and intuitive dashboard interface

- Minimal training required for mosque staff

F. Maintainability

- Easy software/hardware updates
- Modular design for replacing faulty components

G. Cost Efficiency

- Use affordable IoT components suitable for local deployment in Egypt

1.4 Constraints and Assumptions

Constraints

- Limited budget; use cost-effective sensors
- Implementation in Cisco Packet Tracer simulation
- Dependence on internet for prayer-time API

Assumptions

- Stable local network exists in mosque
- Power supply is reliable
- Admin has basic technical knowledge

The system aligns with Egypt's sustainability goals by reducing unnecessary consumption of water and electricity in high-usage public religious spaces.

2. Process Specification

2.1 Overview

The Smart Mosque Management System operates through coordinated processes that continuously collect sensor data, analyze conditions, and trigger automated actions. These processes ensure efficient energy usage, water conservation, and safe occupancy management. The system follows a Sense → Process → Act → Monitor pipeline.

2.2 Core System Processes

2.2.1 Occupancy Detection Process

Objective: Monitor the number of people inside the mosque.

Workflow:

1. PIR sensors at entrances detect entry/exit events.
2. System increments/decrements occupancy count.

3. Data is sent to the central controller.
4. Trigger occupancy-full siren (is-mosque-full-siren) when all hall motion detectors activate simultaneously, and deactivate when any clears.
5. Occupancy data is used by other processes (energy control).

2.2.2 Energy Management Process

Objective: Optimize electricity usage based on prayer times and occupancy.

Workflow:

1. System retrieves prayer times from API.
2. Motion sensors detect presence in prayer hall.
3. Decision logic:
   - IF (Within Prayer Window OR Motion Detected), then turn ON lights and AC/fans.
   - ELSE, turn OFF devices.
4. System also activates AC independently when temperature exceeds 32°C (separate threshold from fans at 28°C). When AC activates, smart window automatically closes. When AC is off and humidity $\geq$ 60%, window opens for passive ventilation.
5. Optionally adjust intensity based on crowd density.

2.2.3 Wudu Water Monitoring Process

Objective: Prevent water waste in ablution areas.

Workflow:

1. Flow sensors measure water usage.
2. System checks for continuous flow without interruption.
3. If abnormal usage is detected:
   - Assume tap left open or leak.
   - Activate water valve to shut water.
4. Log water consumption data.

2.2.4 Air Quality Monitoring Process

Objective: Maintain comfortable and safe air conditions.

Workflow:

1. Sensors read CO_2, humidity, and temperature.
2. Compare readings with predefined thresholds.
3. If air quality is poor, activate ventilation.
4. Humidifier activates when humidity falls below 30% to maintain comfort.

5. Window management provides passive ventilation as a complement to mechanical AC.
6. Update dashboard with real-time values.

2.2.5 Central Monitoring and Control Process

Objective: Provide full system visibility and control.

Workflow:

1. Collect data from all subsystems.
2. Display occupancy, energy usage, water consumption, and air quality on web dashboard.
3. Allow admin override and manual ON/OFF operations.
4. Optionally store historical data.

2.3 Integrated System Flowchart

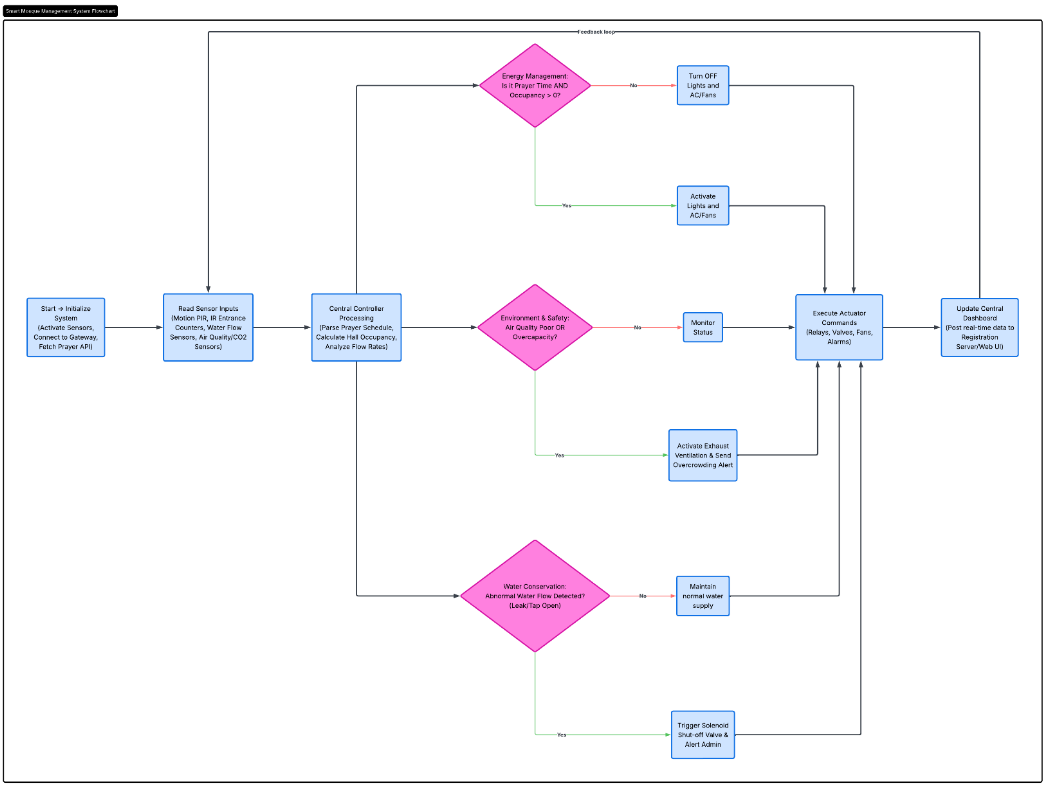


Figure 1: Integrated system flowchart

2.4 Process Interaction Notes

- Occupancy data affects energy decisions.
- Air quality can override energy logic (force ventilation ON).
- Water system is independent but logged centrally.
- All processes communicate via a central controller (MCU/SBC).

The system processes are designed as loosely coupled modules to ensure scalability, maintainability, and real-time responsiveness.

3. Domain Model Specification

3.1 Entities

Core System Entities:

- Mosque
- Zone (Prayer Hall / Wudu Area)
- Smart Window
- Humidifier
- Street Lamp
- Occupancy Siren

Sensors:

- Motion Sensor
- Air Quality Sensor
- Water Flow Sensor

Actuators:

- Light
- Fan / AC
- Water Valve

System Components:

- Controller (MCU/SBC)
- IoT Server
- Dashboard (Web App)
- Admin

3.2 Attributes for each Entity

Mosque

- mosqueID
- name
- location

Zone

- zoneID
- type (Prayer Hall / Wudu Area)
- capacity

Sensor

- sensorID
- type (motion, PIR, air, water)
- status (active/inactive)
- value (reading)

Actuator

- actuatorID
- type (light, fan, valve)
- status (ON/OFF)

Controller

- controllerID
- processingLogic
- status

IoT Server

- serverID
- database
- connectionStatus

Dashboard

- dashboardID
- displayData
- controlOptions

Admin

- adminID
- username
- password

3.3 Relationships

- Mosque contains Zones
- Zone contains Sensors
- Zone contains Actuators
- Sensors send data to Controller
- Controller controls Actuators
- Controller communicates with IoT Server
- IoT Server updates Dashboard
- Admin interacts with Dashboard
- Dashboard sends commands to Controller

3.4 UML Diagram

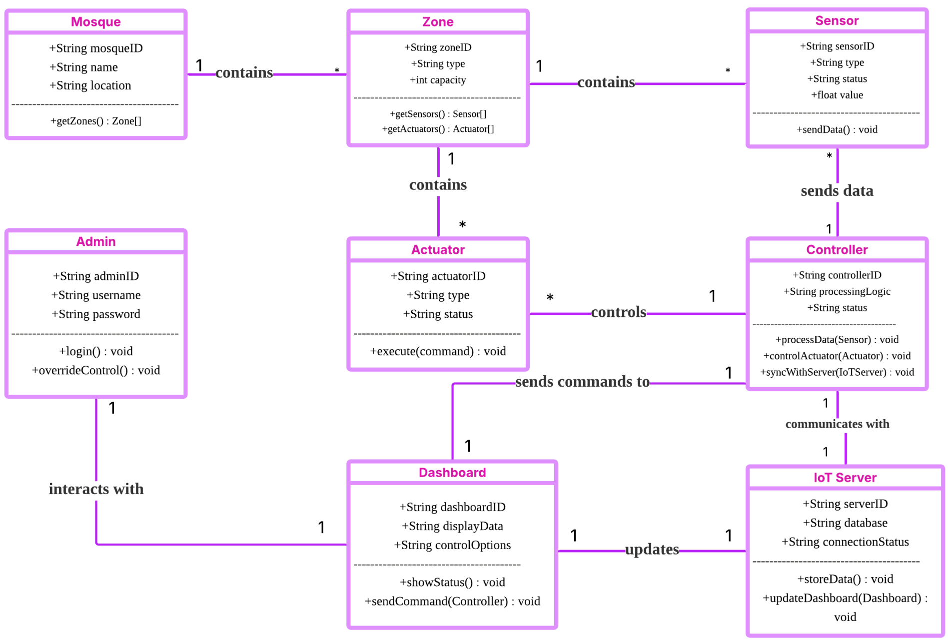


Figure 2: UML domain model diagram

4. Information Model Specification

4.1 Types of Data in System

4.1.1 Sensor Data (Raw Data)

Occupancy Data

- peopleCount (int)
- entry/exit events

Motion Data

- motionDetected (boolean)

Air Quality Data

- CO_2Level (ppm)
- temperature (°C)
- humidity (%)

Water Data

- flowRate (L/min)
- usageDuration (seconds)

4.1.2 Processed Data

- occupancyStatus (normal / full / overcrowded)
- airQualityStatus (good / moderate / poor)
- waterUsageStatus (normal / leak detected)
- energyState (ON / OFF)

4.1.3 System Data (Stored)

- historical water usage
- energy consumption logs
- occupancy logs
- alerts history

4.1.4 Control Data (from Admin)

- manual override commands
- turn ON/OFF devices

4.2 Data Flow

4.2.1 Main Flow

Sensors → Controller → IoT Server → Dashboard

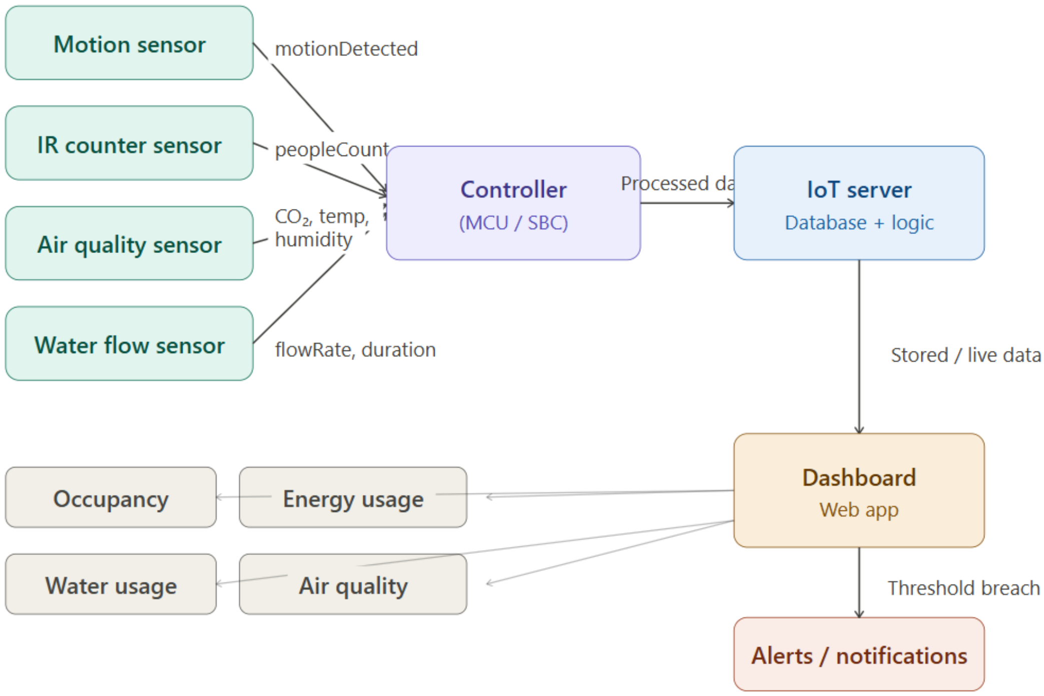


Figure 3: Main data flow

Figure 4: Main flow legend

4.2.2 Control Flow

Admin → Dashboard → Controller → Actuators

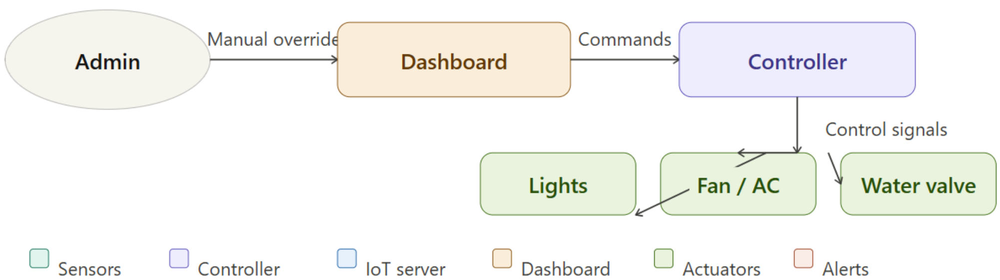


Figure 5: Control flow

5. Service Specifications

5.1 Overview

The Smart Mosque Management System exposes seven core services. Each service is a defined unit of functionality with clear inputs, outputs, triggers, and interfaces. Services are implemented on the ESP32 controller and exposed to the dashboard via the local network.

5.2 Service List

Service ID	Service Name	Type	Trigger
SVC-01	Prayer Time Sync	Data Acquisition	Scheduled – once daily
SVC-02	Occupancy Tracking	Sensing	Continuous (event-driven)
SVC-03	Energy Control	Actuation	Event-driven
SVC-04	Water Monitoring & Control	Sensing + Actuation	Continuous
SVC-05	Air Quality Monitoring	Sensing + Actuation	Periodic (every 60s)
SVC-06	Alert & Notification	Notification	Threshold-triggered
SVC-07	Dashboard Monitoring & Override	Application + Control	On-demand
SVC-08	Window & Ventilation Control	Actuation	Event-driven (AC state + humidity)
SVC-09	Humidity Comfort Control	Sensing + Actuation	Continuous

Table 1: Core service list

5.3 Service Descriptions

SVC-01 – Prayer Time Sync Service

- Purpose: Retrieves accurate daily prayer times to anchor energy automation logic.
- Interface: Daily HTTP GET request to prayer-time API.
- Output: Five daily timestamps (Fajr, Dhuhr, Asr, Maghrib, Isha) stored locally.
- Fallback: Use previous day's cached schedule when API is unreachable.

SVC-02 – Occupancy Tracking Service

- Purpose: Counts people in prayer hall in real time.
- Interface: PIR events → ESP32 counter logic → dashboard polling every 30s.
- Operations: `increment_count()`, `decrement_count()`, `get_current_count()`.
- Output fields: occupancy_count, capacity_percent, status (NORMAL / HIGH / FULL).

SVC-03 – Energy Control Service

- Purpose: Automates lights and fans/AC per zone based on prayer windows and occupancy.
- Interface: ESP32 reads SVC-01 and SVC-02 state, then controls relay GPIOs.
- Decision logic:

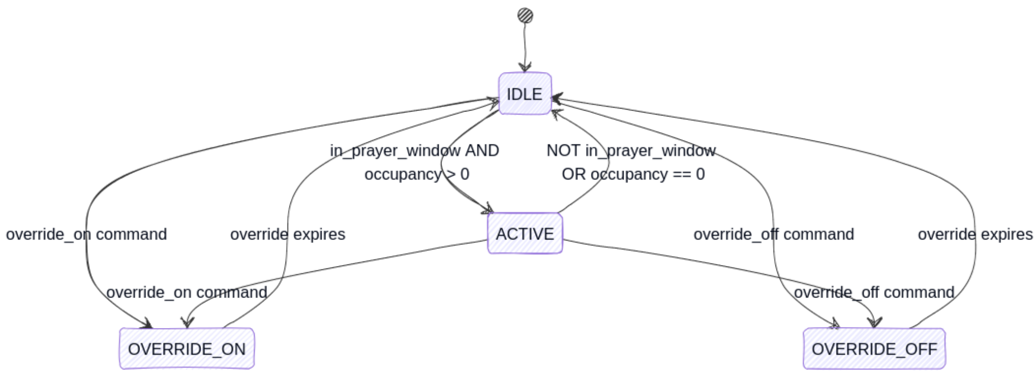


Figure 6: Energy control logic finite state machine

- Air quality from SVC-05 can independently force fan activation.

SVC-04 – Water Monitoring and Control Service

- Purpose: Detect leaks/taps-left-open and control solenoid valve.
- Interface: Flow sensor pulse output → ESP32 pulse counter → valve GPIO.
- Key operations: `monitor_flow()`, `detect_anomaly()`, `close_valve()`, `open_valve()`.
- Failsafe: If flow sensor goes offline, valve defaults to closed and admin is notified.

SVC-05 – Air Quality Monitoring Service

- Purpose: Monitor CO_2, temperature, humidity and auto-activate ventilation.
- Polling interval: Every 60 seconds.

Parameter	Normal	Action Threshold	Action
CO_2	< 1000 ppm	> 1000 ppm	Activate fans
Temperature	18–26 °C	> 28 °C	Activate fans
Humidity	30–60%	> 65%	Activate fans

Table 2: Air quality thresholds

SVC-06 – Alert and Notification Service

- Purpose: Detect threshold breaches and push real-time alerts.
- Interface: Internal event bus, dashboard polls alert queue.
- Deduplication: Same alert type per device/zone can fire once per 5 minutes.

SVC-07 – Dashboard Monitoring and Manual Override Service

- Purpose: Unified monitoring and manual override control.
- Interface: Dashboard polls ESP32 REST endpoints every 30 seconds; override uses HTTP POST.
- Override operations: `override_on()`, `override_off()`, `clear_override()`.
- Safety: Override expires automatically after configured duration.

SVC-08 – Window and Ventilation Control Service

- Purpose: Control smart window behavior to improve ventilation efficiency and reduce unnecessary cooling loss.
- Interface: IoT Server evaluates humidity sensor readings and AC device status, then sends control actions to the smart window device through the wireless access point.
- Decision logic:
  - IF prayer-AC is ON, set prayer-window state to OFF (closed).
  - IF prayer-AC is OFF AND humidity $\geq 60\%$, set prayer-window state to ON (open).
- Trigger source: Event-driven automation rules configured in the IoT Server Conditions page.
- Connected devices: Humidity Monitor, Air Conditioner, Smart Window.

SVC-09 – Humidity Comfort Control Service

- Purpose: Maintain comfortable indoor humidity levels for worshippers.
- Interface: IoT Server continuously monitors humidity readings and controls the humidifier device through configured automation conditions.
- Decision logic:
  - IF humidity $< 30\%$, activate prayer-humidifier.
  - IF humidity $\geq 30\%$, deactivate prayer-humidifier.
- Trigger source: Continuous threshold monitoring configured in the IoT Server.
- Connected devices: Humidity Monitor and Humidifier.

6. IoT Level Specification

The Smart Mosque Management System is structured across five IoT levels, following a standard layered architecture. Each layer has a defined responsibility and communicates with adjacent layers.

6.1 Level 1 – Perception Layer

Role: Direct interaction with the physical environment.

Sensors

Actuators

Device	**Placement**	**Measurement**
PIR Motion Sensor	Main entrance, exit, hall zone	Entry/exit events, zone presence
Water Flow Sensor	Wudu tap inlet pipe	Flow rate (L/min), pulse output
PIR Motion Sensor (Wudu)	Ablution area	Presence near taps
IAQ Sensor	Prayer hall near ventilation	CO_2, temperature, humidity
Humidity Sensor	Prayer hall	Humidity % reading

Table 3: Perception layer sensors

Device	**Placement**	**Controlled Action**
Smart Relay (Lights)	Per hall zone	ON/OFF lighting
Smart Relay (Fans/AC)	Per hall zone, wudu area	ON/OFF ventilation
Solenoid Valve	Wudu tap supply line	Open/close water flow
Smart Window	Prayer hall	Open/close for ventilation
Humidifier	Prayer hall	ON/OFF for humidity comfort
Street Lamp	Mosque entrance	ON/OFF occupancy-based

Table 4: Perception layer actuators

6.2 Level 2 – Network Layer

Role: Local connectivity and data transport.

- Hardware: Local Wi-Fi access point for ESP32 nodes and local server
- Protocol: IEEE 802.11 Wi-Fi with WPA2-PSK
- Topology: Star
- Segmentation: IoT network isolated from guest/user devices

Key decision: Core control loops are local and do not depend on internet connectivity.

6.3 Level 3 – Edge Processing Layer

Role: Real-time control and decision-making.

Hardware: ESP32 microcontrollers per functional zone.

Function	Description
Sensor polling	Reads PIR, IAQ, and flow sensor inputs at configured intervals
Occupancy counting	Increments/decrements people count based on entry/exit events
Prayer window evaluation	Compares current time against cached prayer schedule
Energy control logic	Combines occupancy and prayer window state for relays
Water control logic	Detects anomalies and closes valve when required
Air quality control	Polls IAQ every 60s and activates fans on threshold breach
Alert generation	Publishes threshold breach events to data layer

Table 5: Edge processing responsibilities

Failsafe: If server connectivity is lost, edge controllers continue autonomous operation.

6.4 Level 4 – Data Accumulation Layer

Role: Aggregate processed data, store logs, and serve dashboard queries.

- CPT simulation backend: Registration Server
- Real deployment option: local server (e.g., lightweight REST + database)
- Retention: rolling 30-day window with optional archival

6.5 Level 5 – Application Layer

Role: Human-facing dashboard for monitoring and control.

- Web dashboard on local network
- Single admin role
- Functions: live monitoring, alert display, schedule view, manual override

6.6 Level Interaction Summary

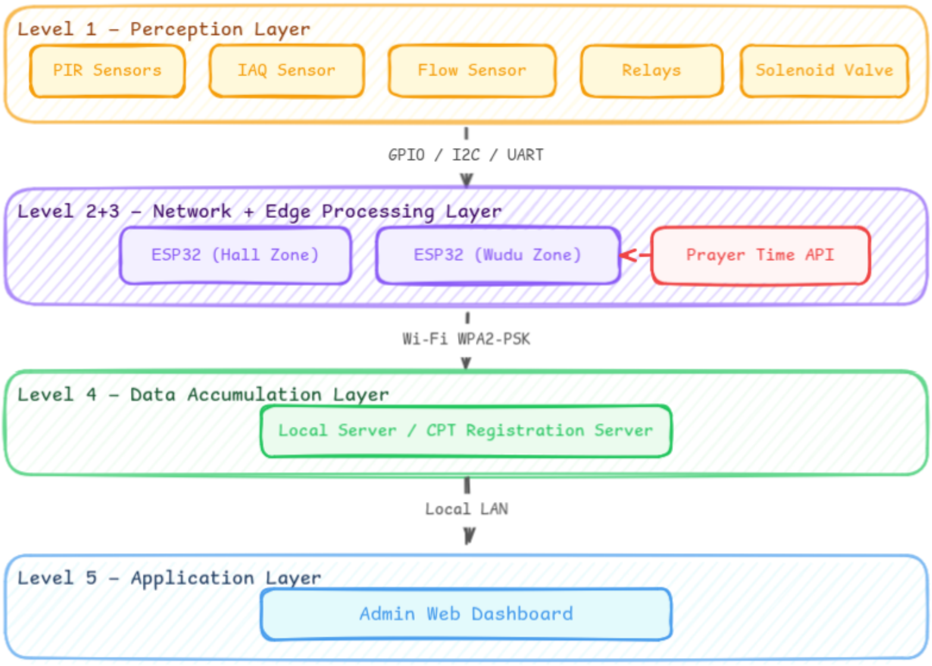


Figure 7: IoT system layer interactions

7. Functional View Specification

7.1 Overview

The functional view breaks the system into independent blocks by behavior. Each block has defined inputs, outputs, and responsibilities, with clear data flows for modularity and maintainability.

7.2 Functional Block Descriptions

Block 1 – Sensing Block

Collects raw data from the mosque environment.

Failure behavior: Offline sensors are flagged to processing, then safe defaults are applied.

Block 2 – Processing Block

Decision core running on ESP32.

- Occupancy counter with floor at zero.
- Prayer window evaluator from cached schedule.
- Energy decision engine for lights/fans.
- Air quality decision engine independent from prayer window.
- Water anomaly detection and valve control.

Sub-function	Sensor	Interval	Output
Entry/exit detection	PIR (entrance)	Event-based	Increment/decrement event
Zone presence detection	PIR (hall zones)	Every 5s	Presence flag per zone
Water flow measurement	Flow sensor	Every 1s	Flow rate (L/min)
Tap-area presence	PIR (wudu)	Event-based	Presence flag
Air quality reading	IAQ sensor	Every 60s	CO_2, temperature, humidity

Table 6: Sensing block functions

- State manager as single source of truth.

Block 3 – Actuation Block

Executes commands on relays and valve, with confirmation and retries.

Device	Command Type	Confirmation
Smart relay (lights)	ON/OFF per zone	GPIO acknowledgment
Smart relay (fans/AC)	ON/OFF per zone	GPIO acknowledgment
Solenoid valve	OPEN/CLOSE per tap	GPIO acknowledgment

Table 7: Actuation block control and feedback

Block 4 – Data and Storage Block

Persists time-series data for dashboard and analytics.

Data Type	Logging Frequency	Retention
Occupancy count	Every 30s	30-day rolling
Water usage per tap	Per session + hourly total	30-day rolling
Device state changes	On every transition	30-day rolling
Air quality readings	Every 60s	30-day rolling
Alert records	On every alert event	30-day rolling

Table 8: Stored operational data

Block 5 – Alert Block

Evaluates alert conditions, deduplicates repeated alerts, and pushes active alerts to dashboard.

Block 6 – Dashboard Block

Provides monitoring panels and manual override controls for admin users.

Block 7 – External API Block

Fetches daily prayer times and caches them for resilience.

7.3 Inter-Block Data Flow Summary

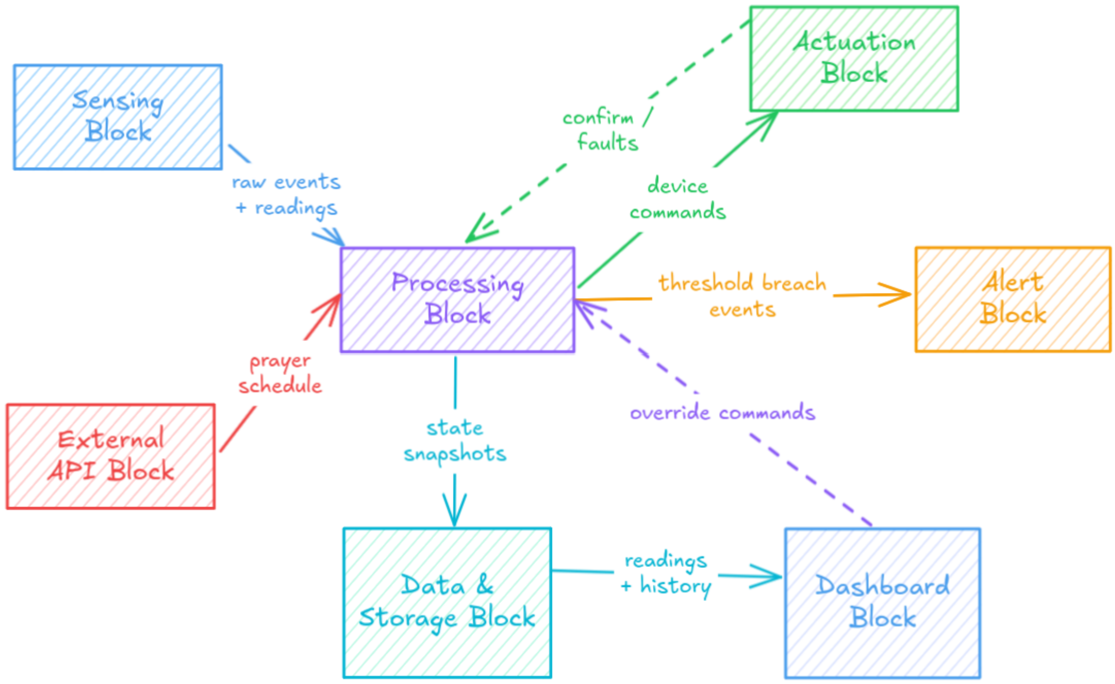


Figure 8: Inter-block data flow

8. Operational View Specification

8.1 System Performance

8.1.1 Response Time

The system operates in near real-time. Motion events trigger lighting and ventilation in approximately 1–2 seconds. In the wudu area, abnormal continuous flow triggers valve closure in approximately 2–3 seconds to reduce waste while limiting false positives.

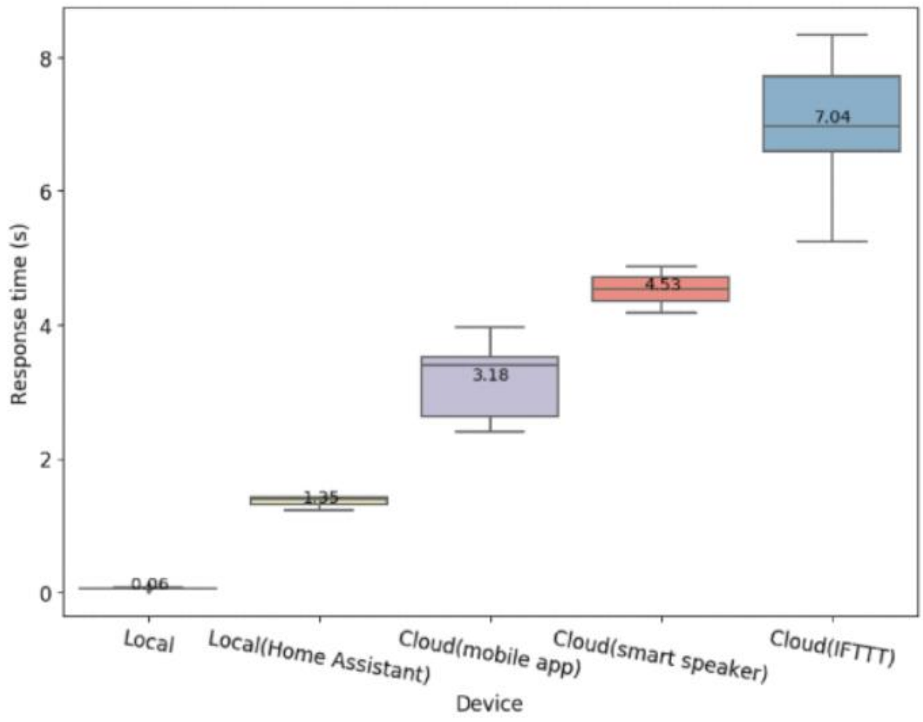


Figure 9: Operational response-time trend

Air quality is monitored periodically and ventilation is adjusted when thresholds are exceeded. Temperature and humidity are also tracked to maintain comfort. The system handles simultaneous sensor inputs during peak times without noticeable degradation.

8.1.2 Data Update Frequency

- Occupancy: real-time update per entry/exit event.
- Air quality: every 1 minute (balanced responsiveness and stability).
- Water usage: continuously monitored in real time.

8.1.3 Throughput

The system processes motion, water-flow, and air-quality streams in parallel, enabling simultaneous monitoring and decision-making during high-demand periods such as Friday prayer and Ramadan.

8.1.4 Efficiency

Lights and fans operate only when needed based on occupancy and schedule, and water control logic closes supply during leak-like behavior to reduce waste.

8.2 System Reliability

8.2.1 Sensor Failure Handling

Fallback mechanisms keep operation safe during sensor faults.

- Motion sensor failure: admin manual override controls lighting and fans.
- Flow sensor failure: manual valve override + admin alert.

- Air-quality sensor failure: ventilation follows safe default schedule.

8.2.2 Network Failure

Core control continues locally if network/server connectivity is lost. Dashboard synchronization resumes automatically after restoration.

8.2.3 Power Failure

On recovery, the system returns to predefined safe startup states. Critical monitoring may be maintained through optional backup power.

8.3 System Scalability

8.3.1 More Sensors

Additional occupancy, flow, and IAQ sensors can be integrated with the same controller logic.

8.3.2 Spreading to Other Mosques

The architecture can be replicated across multiple mosques, each with local operation and data separation.

8.3.3 Modular Architecture

Occupancy, water, air-quality, and dashboard modules can evolve independently, reducing maintenance risk and easing upgrades. The window-ventilation and humidity-comfort modules operate independently and were added without modifying core occupancy or air-quality logic, demonstrating the modular design principle in practice.

8.4 System Security

8.4.1 Network Protection

- Segmented network for IoT devices
- Restricted dashboard/control access
- Firewall/gateway policy enforcement
- Monitoring for suspicious behavior and repeated login attempts

8.4.2 Access Control

Role-based permissions, authenticated overrides, and action logging are applied for accountability.

Authentication flow:

$$\text{User} \rightarrow \text{Login Page} \rightarrow \text{Authentication Check} \rightarrow \text{Access Granted} \rightarrow \text{Dashboard}$$

8.4.3 Data Encryption

WPA2-PSK secures Wi-Fi communication between sensors, controllers, and dashboard. AES encryption protects traffic confidentiality and integrity.

8.4.4 Device Filtering

MAC allow-list filtering blocks unknown devices from joining. It is used together with WPA2-PSK and access control (not standalone security).

8.4.5 Privacy Protection

The system stores aggregate occupancy counts only and avoids personal identification. PIR-based sensing does not capture images.

8.5 Energy Efficiency

8.5.1 Prayer Time Control

Prayer-time-aware control avoids continuous operation outside worship periods.

8.5.2 Occupancy-Based Activation

Devices activate only when people are present, reducing unnecessary consumption.

8.5.3 Automatic Shutdown

Lights and fans turn off after inactivity timeout, minimizing waste.

8.6 Safety Considerations

8.6.1 Device Overheating

Thermal-safe operation requires ventilation, safe power design, and alerts/shutdown under abnormal heating.

8.6.2 Water Valve Reliability

Valve closure is validated over a short window to avoid false shutdown during normal use interruptions.

9. Device and Component Integration

9.1 Input Devices (Sensors)

9.1.1 Motion Sensor (PIR)

A Passive Infrared (PIR) sensor detects changes in infrared radiation emitted by warm bodies. It is compact, low-cost, and energy-efficient.

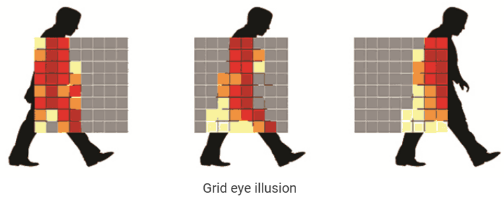


Figure 10: PIR motion sensor (left) and PIR sensing grid illustration (right)

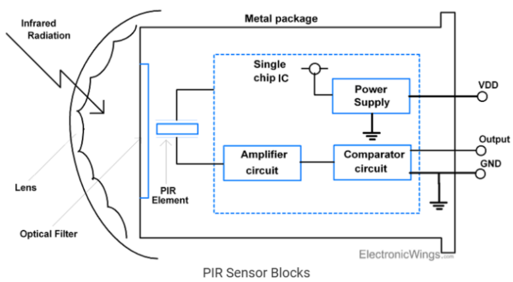
Sensor details:

- Detection range: up to 7 meters
- Detection angle: 110 degrees
- Operating voltage: DC 4.5V–12V
- Output signal: 3.3V digital output
- Delay time: 0.3 seconds to 5 minutes (adjustable)
- Operating temperature: -15°C to +70°C
- Sensitivity: adjustable

Working principle summary:

1. Infrared detection: The PIR sensing element continuously detects infrared radiation from the environment. When a warm object crosses the field of view, the detected radiation level changes from one zone to another, producing a measurable differential signal. This transition is the key trigger used by the sensor to recognize motion.

2. Sensor structure: Uses dual sensing zones and lens focusing.



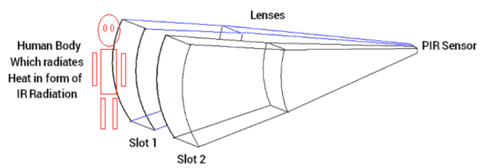


Figure 11: Infrared detection concept (left) and PIR sensor structure (right)

3. Idle state: Balanced sensing zones keep the output stable.

4. Motion detection: As a person moves across the sensing area, one PIR segment receives higher infrared energy than the other. The sensor circuit amplifies this difference and interprets it as movement.

5. Output generation: After threshold comparison, the sensor produces a digital output pulse. This output is forwarded to the controller to trigger occupancy and automation logic in real time.

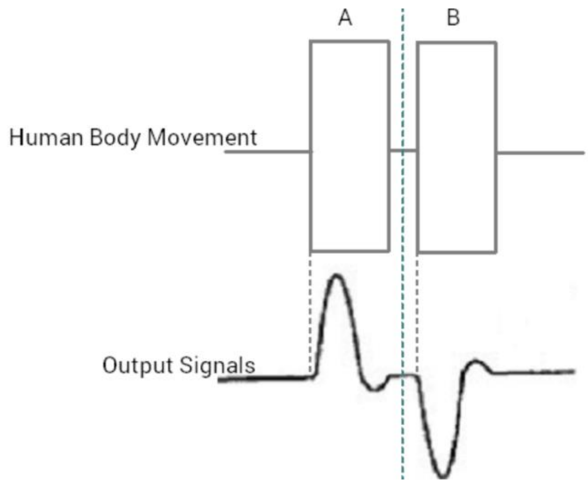


Figure 12: Motion detection behavior

PIR should be installed away from Wi-Fi antennas to reduce electromagnetic interference risk.

9.1.2 Air Quality Sensor (IAQ)

Figure 13: Indoor air quality sensor module

IAQ sensing supports ventilation control by monitoring gases and climate parameters.

Key IAQ factors:

- Chemical pollutants (CO_2, VOCs, NO_2)
- Biological contaminants (mold, bacteria, allergens)
- Particulate matter (PM2.5, PM10)
- Environmental conditions (humidity, temperature, pressure)

Common pollutants and impact:

Pollutant	Description	Health Effects
CO_2	Ventilation indicator	Fatigue, headache, lower productivity
VOCs	Emitted by paints/cleaners/plastics	Irritation, dizziness, long-term risk
PM2.5/PM10	Fine dust and smoke particles	Respiratory and cardiovascular burden
CO	Incomplete combustion product	Severe toxicity at high concentration
Formaldehyde	Emitted by furniture/insulation	Carcinogenic, irritation
NO_2/O_3	Combustion and outdoor sources	Respiratory distress
Humidity	Moisture level	Mold growth, discomfort
Temperature	Thermal condition	Fatigue, reduced comfort

Table 9: IAQ factors and effects

IAQ sensors require periodic calibration (typically every 6–12 months).

9.1.3 Water Flow Sensor

Figure 14: Water flow sensor

Water flow sensors measure instantaneous flow rate and cumulative usage, producing pulse output suitable for MCU integration. This allows the controller to detect abnormal continuous flow and apply water-saving actions when needed.

Working principle summary:

- Water drives an internal rotor.
- Hall-effect sensing converts rotor speed to pulse signal.
- Pulse frequency maps to flow rate.
- Controller computes usage and triggers control logic.

9.2 Output Devices (Actuators)

9.2.1 Smart Lights

Activated by combined prayer-window and occupancy conditions to balance comfort and energy savings.

9.2.2 Fans (AC Simulation)

Activated by IAQ conditions and occupancy demand. Two separate cooling thresholds are implemented:

- Ventilation fan activates when CO_2 exceeds threshold, temperature exceeds 28°C, or humidity exceeds 65%.
- Dedicated AC unit activates when temperature exceeds 32°C as a stronger cooling response.

When environmental conditions normalize, the system automatically deactivates the corresponding cooling devices.

9.2.3 Solenoid Valve

Automatically closes water supply during abnormal flow patterns and can reopen via logic/manual reset.

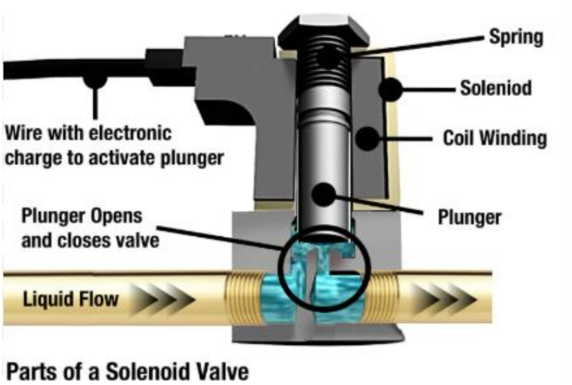


Figure 15: Solenoid valve components

9.2.4 Smart Window

The smart window is controlled automatically using AC state and humidity readings. When the AC unit activates, the window closes to preserve conditioned air inside the prayer hall. If the AC is OFF and humidity reaches high levels, the window opens to support passive ventilation and improve indoor airflow.

9.2.5 Humidifier

The humidifier activates automatically when indoor humidity drops below the configured comfort threshold. This helps maintain comfortable environmental conditions for worshippers, especially during dry weather conditions.

9.2.6 Street Lamp

The street lamp provides entrance-area lighting based on motion detection and occupancy activity near the mosque entrance. This improves visibility and safety while reducing unnecessary energy consumption during low-activity periods.

9.3 Processing Unit

The ESP32 is selected for integrated Wi-Fi/Bluetooth, sufficient GPIO and peripheral interfaces (I2C/UART/SPI/PWM), low power operation, and capability to run concurrent IoT control tasks.

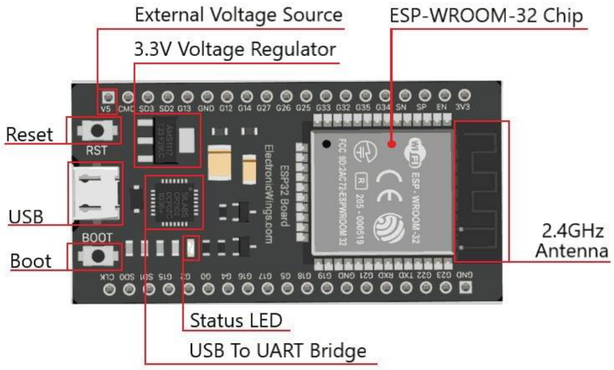


Figure 16: ESP32 processing unit

9.4 Communication Layer

- Local Wi-Fi AP for ESP32-to-server connectivity
- IEEE 802.11 with WPA2-PSK security
- Star topology
- IoT network segmentation from general user devices
- CPT representation: LAN linking MCU units to Registration Server

9.5 Integration Workflow

1. Sensing stage
2. Data transmission stage
3. Processing and decision stage
4. Actuation stage
5. Monitoring and feedback stage

This workflow is consistent with Step 2 process design.

9.6 System Diagram

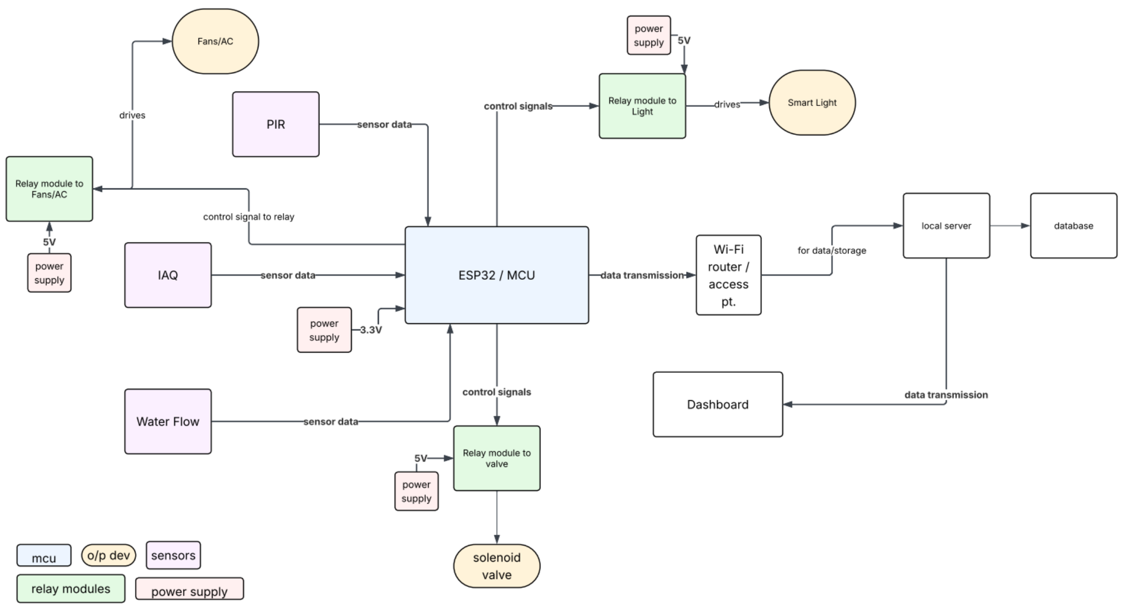


Figure 17: System integration diagram

10. Application Development

10.1 Application Overview and Platform

The application layer is a web-based dashboard acting as the primary interface between users and the IoT system. It centralizes real-time data visualization and device control in one operational view.

10.2 User Role

The system is designed for an admin role with full access to monitoring, alert handling, and override control to maintain system performance and safety.

10.3 Core Features

- Real-time monitoring (CO_2, temperature, humidity, occupancy)
- Live status display of connected devices
- Alert and notification handling for abnormal conditions
- Actuator control (e.g., ventilation)
- Historical data overview and basic trend visualization

Application-system interaction:

- Retrieves real-time sensor data via backend APIs
- Displays continuously updated system status
- Sends control commands to actuators

- Supports bidirectional monitoring/control communication
- Maintains low-latency updates for operational responsiveness

10.4 User Interface Design

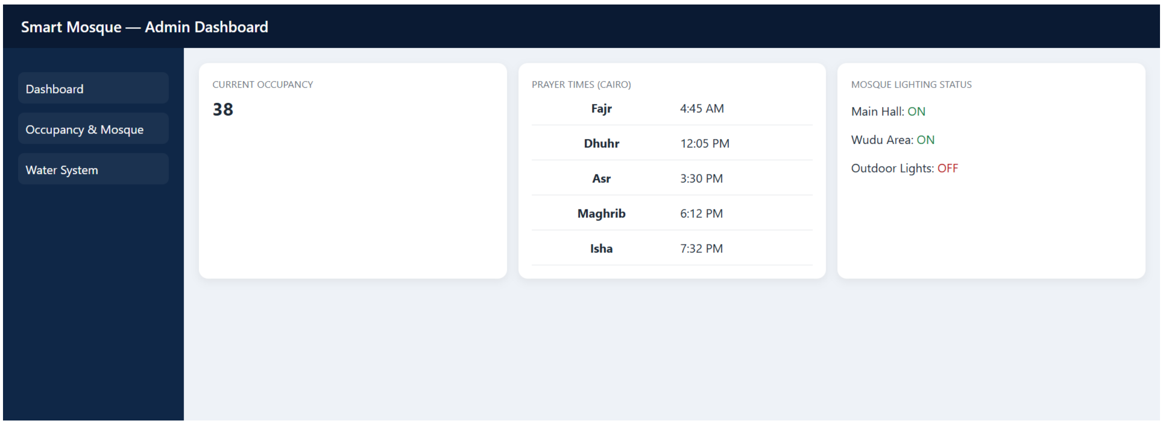


Figure 18: Dashboard UI design - Prayer times

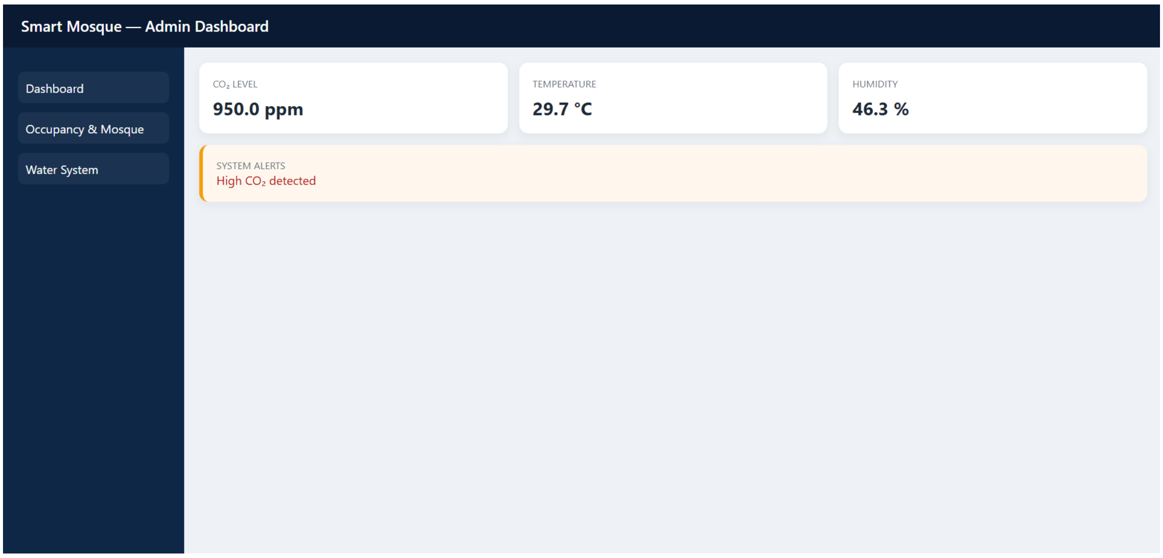


Figure 19: Dashboard UI design - High CO_2 alert

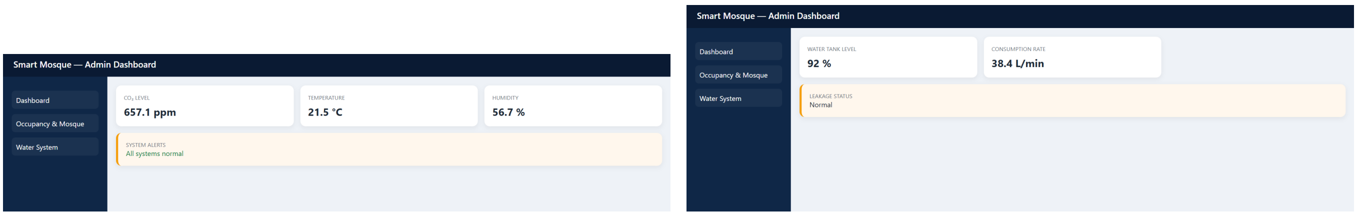


Figure 20: Dashboard UI design - Normal conditions

List of Figures

References

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